Proof of Concept (PoC)

File Integrity Monitoring (FIM) Using Wazuh

Researcher Information:

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Title:

File Integrity Monitoring and Log Analysis Using Wazuh

Description:

This PoC demonstrates how File Integrity Monitoring (FIM) detects unauthorized file changes using **Wazuh**.

The lab includes:

- Installing Wazuh Manager and Agent
- Enabling File Integrity Monitoring
- Generating file changes
- Monitoring alerts in dashboard

Environment:

Monitoring Tool: Wazuh
System: Linux (Kali / Ubuntu)
Setup: Local Lab Environment

Tools Used

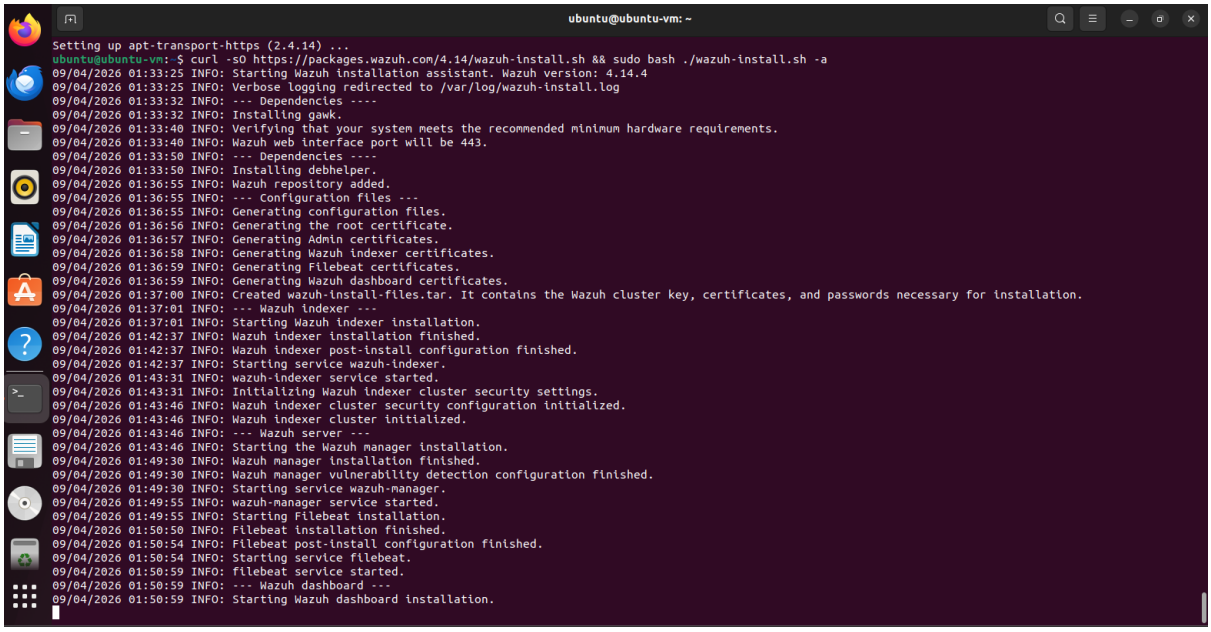
- Wazuh Manager
- Wazuh Agent
- Wazuh Dashboard

Step 1: Install Wazuh Manager

Install Wazuh manager on the server system.

Purpose:

- Central system for monitoring and alerts



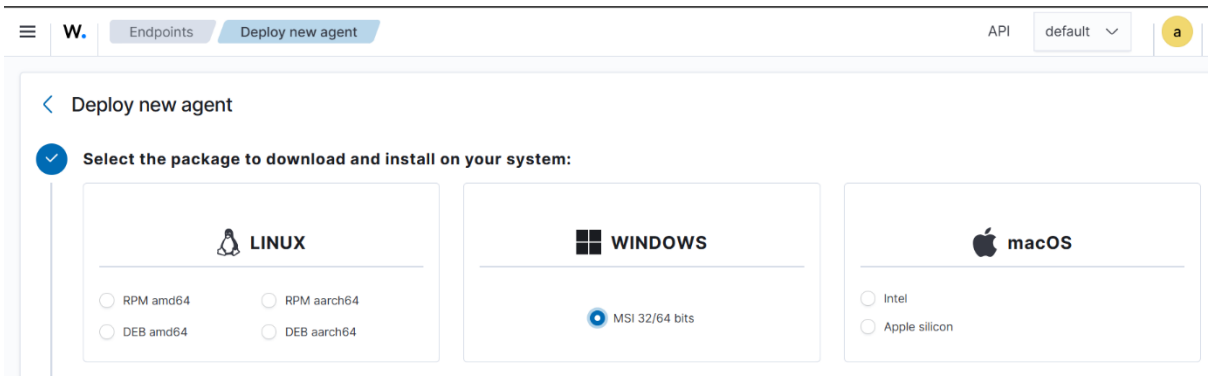
Wazuh manager installation completed

Step 2: Install and Connect Wazuh Agent

Install agent on monitored system and connect to manager.

Purpose:

- Send logs and file changes to manager



Agent successfully connected to manager

Step 3: Enable File Integrity Monitoring (FIM)

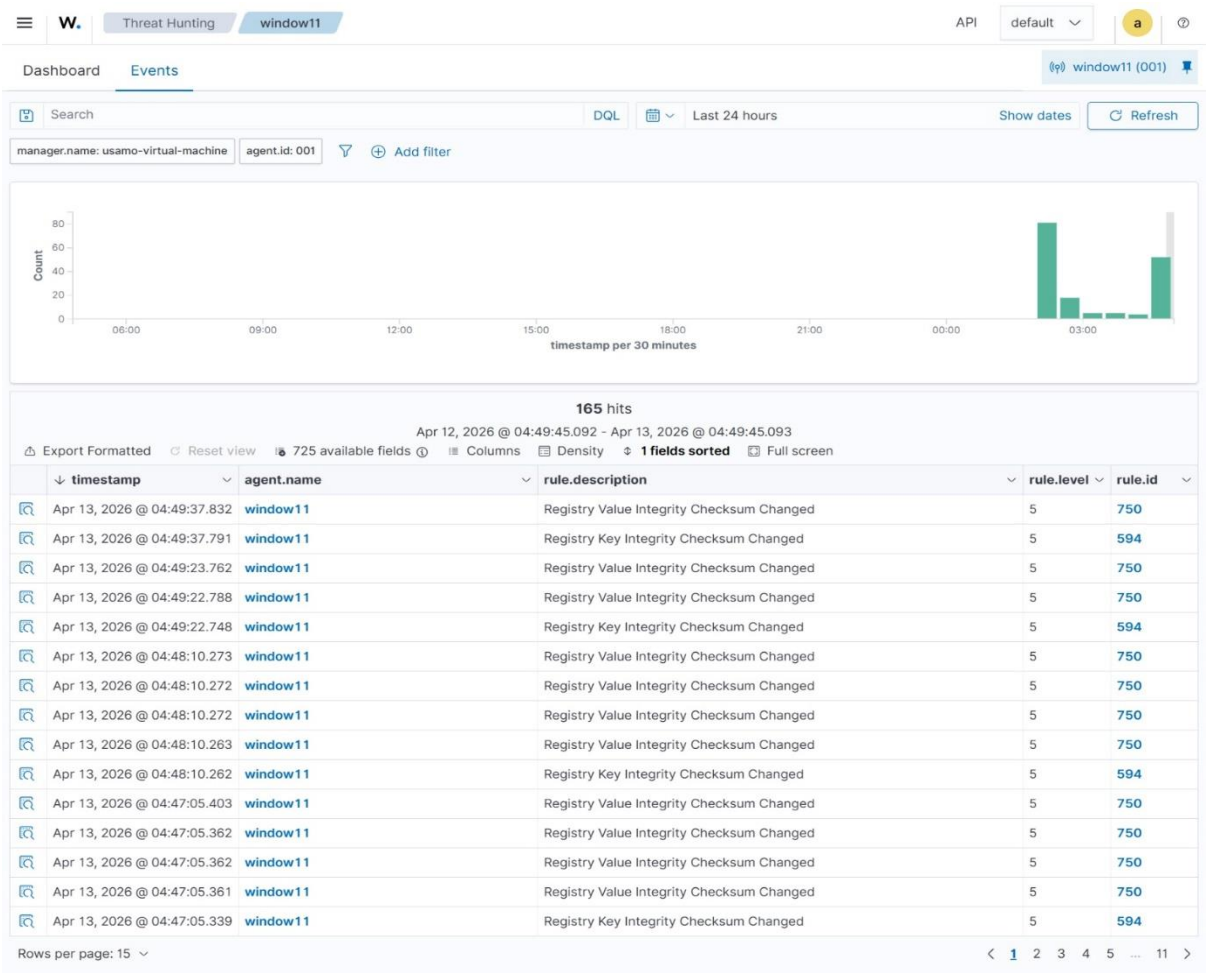
Configure FIM in Wazuh configuration file.

Example path:

```
/var/ossec/etc/ossec.conf
```

Purpose:

- Monitor critical directories for changes



timestamp	agent.name	rule.description	rule.level	rule.id
Apr 13, 2026 @ 04:49:37.832	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:49:37.791	window11	Registry Key Integrity Checksum Changed	5	594
Apr 13, 2026 @ 04:49:23.762	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:49:22.788	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:49:22.748	window11	Registry Key Integrity Checksum Changed	5	594
Apr 13, 2026 @ 04:48:10.273	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:48:10.272	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:48:10.272	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:48:10.263	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:48:10.262	window11	Registry Key Integrity Checksum Changed	5	594
Apr 13, 2026 @ 04:47:05.403	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:47:05.362	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:47:05.362	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:47:05.361	window11	Registry Value Integrity Checksum Changed	5	750
Apr 13, 2026 @ 04:47:05.339	window11	Registry Key Integrity Checksum Changed	5	594

FIM configuration enabled (show monitored directory)

Step 4: Restart Wazuh Services

Apply configuration changes.

Command:

```
sudo systemctl restart wazuh-manager
```

Purpose:

- Activate FIM monitoring

Step 5: Generate File Changes (Test)

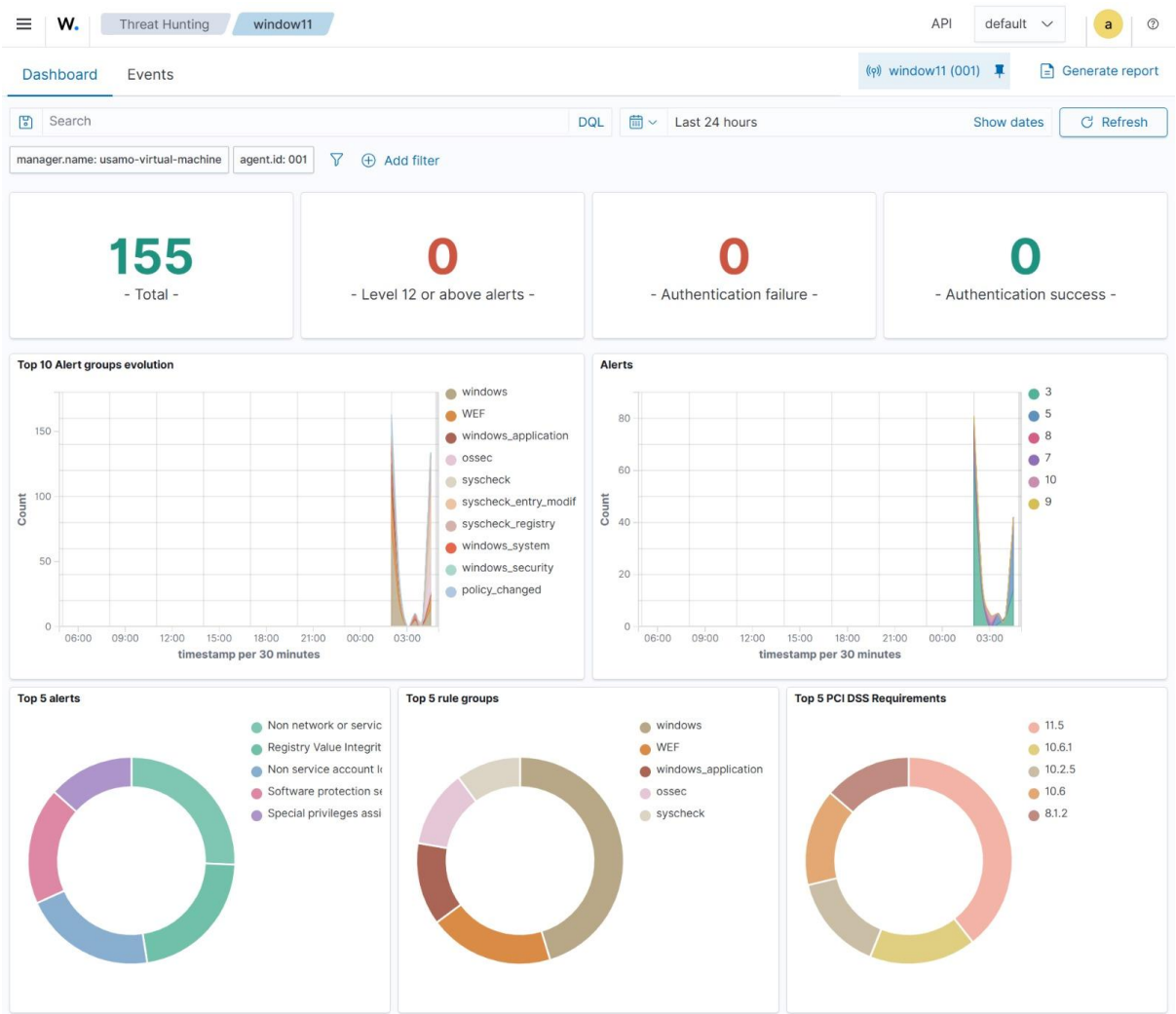
Create or modify a file in monitored directory.

Example:

```
echo "test" >> testfile.txt
```

Purpose:

- Trigger FIM alert



File created/modified

Step 6: Monitor Alerts in Wazuh Dashboard

Open Wazuh dashboard and check alerts.

Navigate to:

- Security Events
- File Integrity Monitoring

Purpose:

- Detect unauthorized changes

FIM alert showing file modification

Step 7: Analyze Logs

Review details of alert:

- File name
- Time of change
- Action (created/modified/deleted)

Proof of Monitoring

- File change detected successfully
- Alert generated in dashboard
- Logs show file activity

Key Findings:

- Wazuh successfully detected file changes
- Alerts generated in real-time
- Monitoring works for unauthorized modifications

Impact:

Without FIM:

- File tampering goes unnoticed
- Attackers can modify system files
- No visibility into system changes

Remediation:

- Enable FIM on critical directories
- Monitor logs regularly
- Configure alerting system
- Integrate with SIEM for better visibility

Conclusion:

This PoC demonstrates that Wazuh File Integrity Monitoring can effectively detect file changes and provide real-time alerts, helping in early detection of suspicious activities.